

trigger key > /?value1 = 1

For Light Off, the link is <https://maker.ifttt.com/trigger/<your event name>/with/key/<your webhook trigger key>/?value1 = 0>

name > /with/key/ < your webhook trigger key > /?value1 = 0

Copy and save these links for

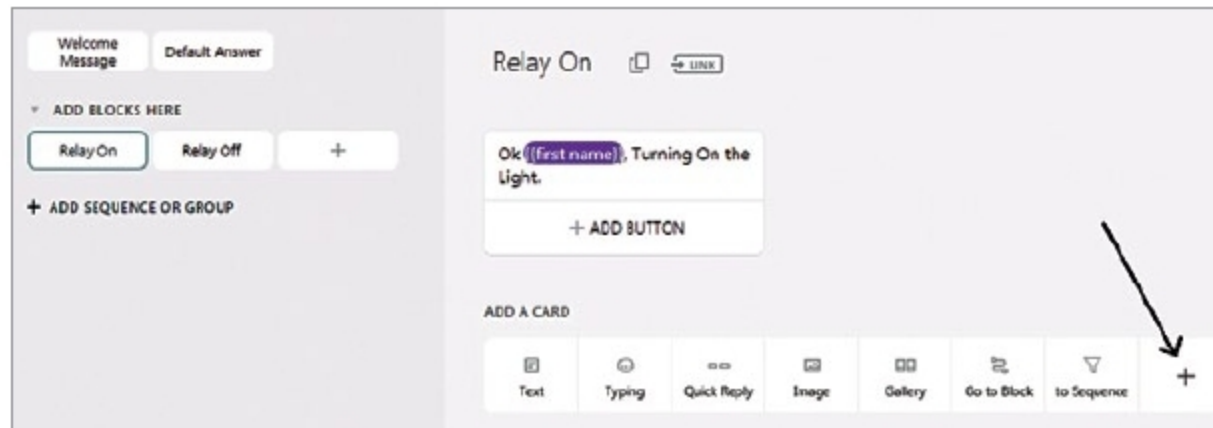


Fig. 21: Add link on Relay On block

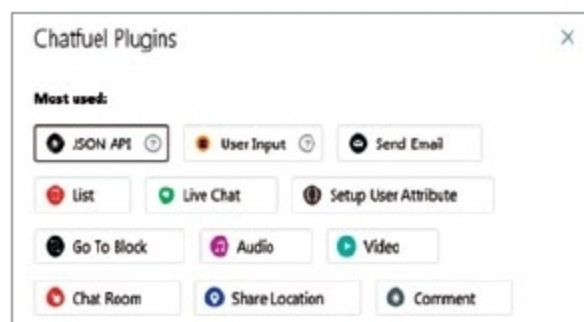


Fig. 22: JSON API in Chatfuel Plugins

Chatfuel integration. Test the links by populating these in the Web browser and checking Toggle Switch created at Adafruit IO server.

Let us integrate everything now.

Open Relay On and Relay Off blocks created in Chatfuel platform. Add JSON APIs to these. Select + sign in Add a Card menu of Relay On block.

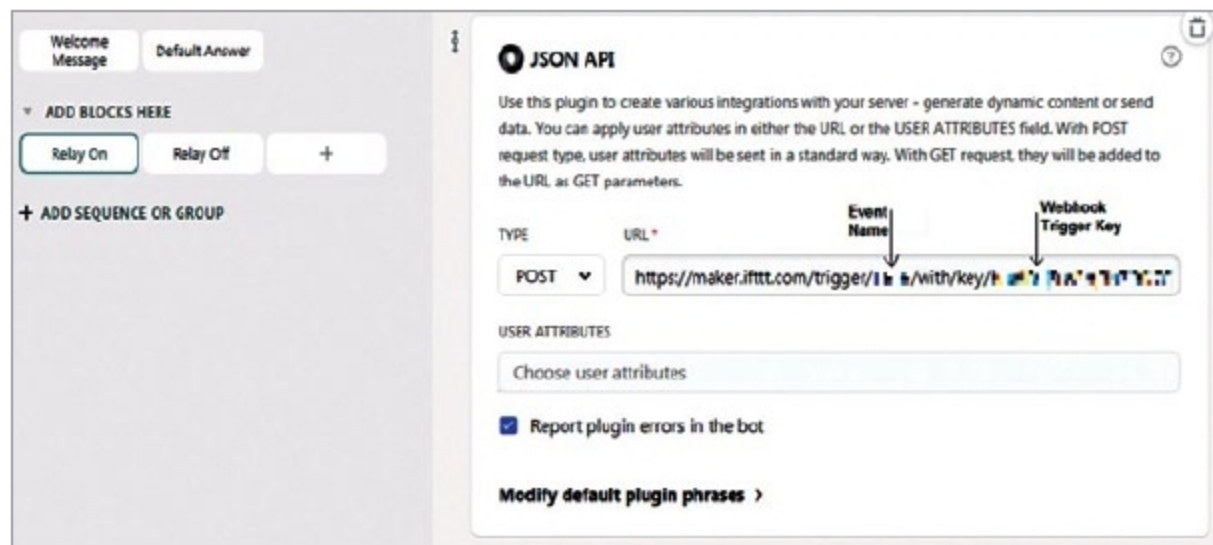


Fig. 23: JSON API of Relay On

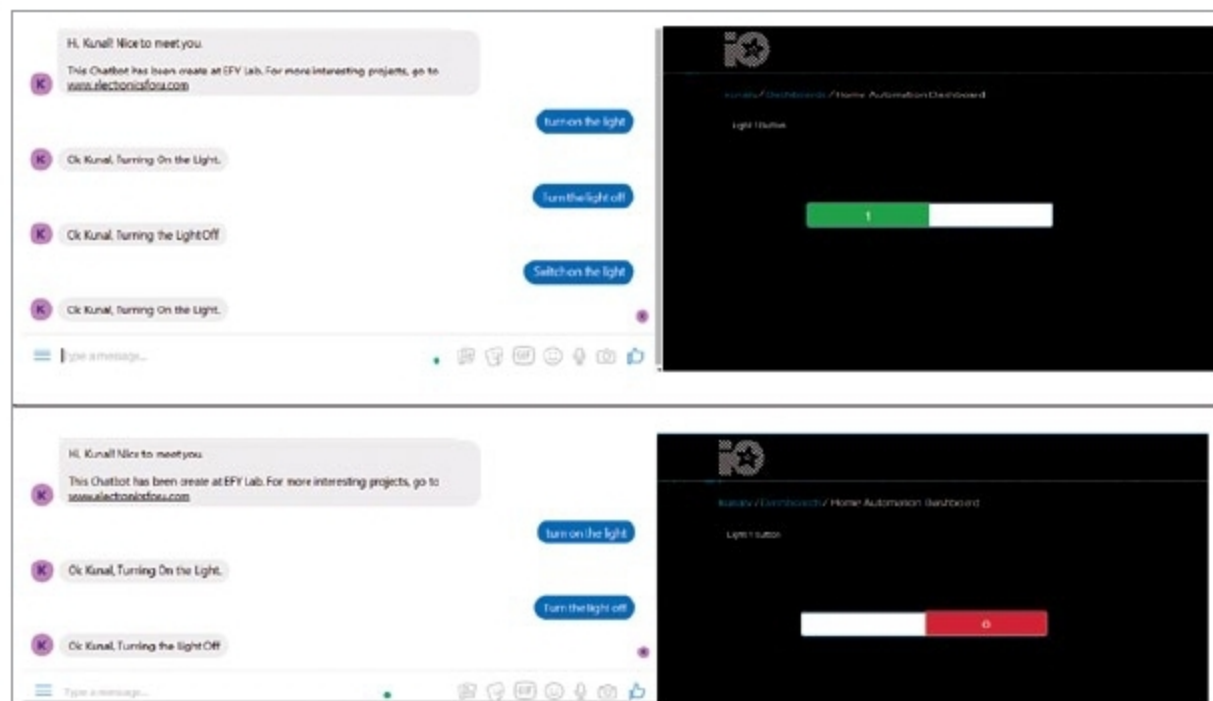


Fig. 24: Test chatbot with Adafruit IO

Click on JSON API from Chatfuel Plugins menu (Fig. 22). Select type as Post and enter the link created for Light On. Paste it in the URL Tab, as shown in Fig. 23.

Similarly, do this for Relay Off.

The chatbot is ready and is connected to the IoT server (Adafruit). Test the chatbot with Inbox section of Chatfuel. Adafruit IO server will get updated in real time, as shown in Fig. 24.

Arduino setup

NodeMCU requires Arduino integrated development environment (IDE) software. Arduino IDE is a cross-platform application. The latest Arduino IDE is available at www.arduino.cc/en/Main/Software

Arduino IDE does not contain support for MQTT and ESP8266 family. Adafruit_MQTT library is available at github.com. To install ESP8266 board library in Arduino IDE, follow the steps given below.

1. Open Arduino IDE and select File > Preferences.

2. Enter http://arduino.esp8266.com/stable/package_esp8266com_index.json in Additional Board Manager URLs field. Click OK.

3. Go to Tools > Board > Boards Manager.

4. Scroll down and select ESP8266 board menu. Install ESP8266 by ESP8266 community.

5. Select your ESP8266 board from Tools > NodeMCU 1.0 (ESP12E module).

6. Restart your Arduino IDE.

Software code (facebook_chatbot.ino) is written in Arduino programming language. Before compiling and uploading to NodeMCU board, make some changes in the code.

Place your Wi-Fi credentials as indicated within double quotes shown below in code declaration. ESP8266WiFi.h library helps run the functions of Wi-Fi in Arduino IDE.

```
#include <Adafruit_MQTT.h> //Adafruit
                                MQTT Libraries
#include <Adafruit_MQTT_Client.h> //
#include <ESP8266WiFi.h> //ESP8266 Wi-Fi
                                libraries
#define relayPin D3 //declare the relay pin
#define WLAN_SSID "xxxxxxx" //enter your
WiFi network name within the double quotes &
```